E-commerce Transactions

Data Cleaning, EDA, Visualization & Insights

Dataset 1: E-commerce Transactions

Task 1: Dataset Understanding

1. Load the dataset into SQL

I imported the OnlineRetail.csv file into SQL Server as a table called OnlineRetail.

2. First few rows

The table has 8 columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

3. Identifying the structure

Rows and columns: The dataset has 541,907 rows and 8 columns.
Data types: InvoiceNo, StockCode, Description, and Country are text (nvarchar). Quantity and CustomerID are whole numbers (int). UnitPrice is a decimal number (float). InvoiceDate is a date/time field (datetime2).
Numerical features: Quantity and UnitPrice — these are the columns that actually make sense to do math with, like totals or averages.
Categorical features: InvoiceNo, StockCode, Description, CustomerID, and Country. CustomerID is stored as a number, but it's really just an ID label for a person — you wouldn't average two CustomerIDs together so it belongs here, not with the numerical features.
Date feature: InvoiceDate.
Possible primary key: No single column uniquely identifies each row. InvoiceNo repeats heavily (541,909 rows but only 25,900 unique invoices), since one order usually contains several products. I tested InvoiceNo + StockCode together as a possible composite key, but found 9,696 combinations that still repeat. Looking at actual examples, some of these are true exact duplicates (every column identical), while others show the same product on the same invoice with a different quantity, which may represent separate entries rather than errors. A clean primary key should be possible once true duplicates are removed during cleaning.

4. What the dataset contains

This dataset is a transaction-level sales log from a UK-based online retail store, covering purchases from December 2010 onward. Each row represents one product bought as part of a customer's order, including the quantity purchased, price per unit, date and time of purchase, and the customer's country. Most orders come from the United Kingdom, but the store also has customers from other countries.

Task 2: Data Cleaning

Missing Values

I checked every column for missing values using a single query that counts blanks per column. Out of all 8 columns, only 3 had missing data:

Column	Missing Count	Method Used
UnitPrice	2	Removed rows

Description	1,454	Filled with "Unknown"
CustomerID	135,080	Left as NULL

UnitPrice — Missing Values
Only 2 rows out of 541,909 had no price listed. UnitPrice is a numerical column used for real calculations like total revenue, so I couldn't safely fill it with a placeholder number without distorting those totals. Since the number of affected rows was so small, I removed them instead of trying to guess a price.

Description — Missing Values
1,454 rows had no product description. Since Description is a text column and doesn't get used in calculations, I filled the blanks with "Unknown" instead of deleting the rows. This kept the row's other valid data (quantity, price, invoice number) usable for revenue and quantity analysis, while being honest that the product name wasn't available.

CustomerID — Missing Values
135,080 rows had no CustomerID — almost 1 in every 4 rows. I initially considered filling these with 0 as a placeholder, but decided against it: 0 isn't a real customer ID, and someone analyzing the data later could mistake it for an actual customer. I left these values as NULL instead, since NULL already represents "unknown" correctly in SQL without inventing a fake ID. CustomerID also isn't needed for most of the analysis in this project (revenue, sales trends, product performance) — it only matters for customer-specific analysis, which can simply filter these rows out using WHERE CustomerID IS NOT NULL when needed.

Duplicate Records

I checked for duplicate rows by grouping on all 8 columns together and looking for any combination that appeared more than once. This found 4,879 duplicate groups — rows that were fully identical across every column.
To remove them, I tagged every row with a row number within its duplicate group (using ROW_NUMBER), keeping the first occurrence as row number 1, and labeling any repeats as 2, 3, and so on. I then deleted every row where the row number was greater than 1, keeping just one copy of each group.

Result: The table went from 541,907 rows down to 536,639 rows — a total of 5,268 duplicate rows removed. This is higher than the 4,879 duplicate groups identified because 291 of those groups had three or more identical copies rather than just two, so more than one extra row had to be removed from those specific groups.

Standardization

Date Formats: Checked the InvoiceDate column and found it was already consistent and correctly stored as a proper date/time type (YYYY-MM-DD HH:MM:SS) for every row. No fixes were needed here.
Text Formatting: Checked the Description column by sorting it alphabetically and found mixed casing (e.g., "*Boombox Ipod Classic" instead of the standard all-caps format used for real

products). Converted the entire column to uppercase so all product names follow one consistent format.
Column Names: Checked all column names for consistency. All columns follow the same naming style (PascalCase, no spaces or special characters). I also noticed a helper column, RowNum, which I had added earlier purely to identify and remove duplicate rows. Since it wasn't part of the original dataset, I dropped it once its job was done, bringing the table back down to a clean 8 columns.
Data Types: Reviewed all column data types. All types are appropriate: text columns stored as nvarchar, Quantity and CustomerID as whole numbers (int), UnitPrice as a decimal (float), and InvoiceDate as a proper date/time field (datetime2). No changes were needed here.

Data Validation

Invalid Numeric Values
I checked Quantity and UnitPrice for values that didn't make sense for a normal sale. I found 10,587 rows with a negative Quantity, and 2,510 rows with a UnitPrice of zero or less.
Looking closer at the zero/negative price rows, nearly all of them had CustomerID as NULL and Description as "UNKNOWN" (a couple were labeled "AMAZON"). This pattern indicated these were not real customer sales, but likely internal stock adjustments or transfers. Since the goal of this project is to analyze sales performance, keeping zero-priced, unidentified transactions would have distorted that analysis. I removed these 2,510 rows from the cleaned dataset.
The 10,587 negative Quantity rows were identified but retained, since they most likely represent legitimate order cancellations or returns rather than data errors.

Inconsistent Records
I reviewed the Country column and found it consistent in casing and format, but noticed three entries used abbreviations or old names instead of full country names: "EIRE" (old name for Ireland), "RSA" (South Africa), and "USA" (United States). Since every other country in the dataset uses its full common name, I standardized these three for clarity and consistency, updating them to "Ireland," "South Africa," and "United States" respectively.

Invalid Text Values
Earlier in the cleaning process (before removing the zero/negative price rows), I had identified placeholder or note-like text in the Description column — entries like "?", "??", "???LOST", and "????MISSING" instead of real product names. After removing the invalid zero/negative price rows, I re-checked this column and found no junk text values remaining, since these entries had overlapped with the zero-priced, non-sale records already removed in the previous step.

Incorrect Formats
Reviewed InvoiceNo and StockCode for consistent formatting. Both generally follow a consistent alphanumeric pattern, with no broken or malformed values observed.

Outliers/Anomalies
The Quantity column was reviewed to identify potential outliers. The two highest values (80,995 and 74,215 units) stood out as significantly larger than the rest of the data — roughly 20 times higher than the next highest order. However, both records had valid product descriptions and

normal, non-zero unit prices, suggesting these were likely legitimate large bulk purchases rather than data entry errors. No changes were made to these records, though they are noted here as notable outliers.
Separately, when reviewing UnitPrice outliers, I found some very high values (up to £38,970) came from non-product entries like "MANUAL," "AMAZON FEE," and "ADJUST BAD DEBT" — internal accounting/administrative records rather than real customer sales. These 596 rows were removed, since they would have distorted revenue figures in the analysis stage.

Cleaning Summary

Issue Found	Action Taken
Missing Values	Description filled with "Unknown" (1,454 rows); CustomerID left as NULL (135,080 rows); UnitPrice rows removed (2 rows)
Duplicates	Removed (5,268 duplicate rows)
Invalid Entries	Corrected — removed 2,510 zero/negative price rows and 596 non-product admin entries (MANUAL, AMAZON FEE, ADJUST BAD DEBT); fixed inconsistent country names (EIRE, RSA, USA)
Standardization	Applied — Description converted to uppercase; date formats, column names, and data types confirmed already consistent

Task 3: Exploratory Data Analysis (EDA)

Summary Statistics

Statistic	Quantity	UnitPrice
Mean	9	£4.70
Median	3	£2.10
Minimum	-80,995	£0.001
Maximum	80,995	£38,970
Standard Deviation	216.45	95.08

The mean is noticeably higher than the median for both Quantity (9 vs 3) and UnitPrice (£4.70 vs £2.10). This shows both columns are right-skewed — most transactions involve small quantities and everyday prices, but a handful of extreme outliers (like very large bulk orders, and a few high-value administrative entries) pull the average upward.

Exploratory Analysis 1: Top-Selling Products (by Quantity)
The best-selling product by total units sold was "PAPER CRAFT, LITTLE BIRDIE" with 80,995 units, followed by "MEDIUM CERAMIC TOP STORAGE JAR" (78,033) and "WORLD WAR 2 GLIDERS ASSTD DESIGNS" (54,951). The top sellers are a mix of home décor and novelty items, consistent with the store's overall product range.

Exploratory Analysis 2: Revenue by Country

The United Kingdom generated by far the most revenue, at approximately £9,001,744 — about 31 times more than the next highest country, the Netherlands (£285,446). Ireland, Germany, and France rounded out the top 5. This heavy concentration confirms the store's primary market is domestic (UK-based), with international sales making up a relatively small share of total revenue.

Exploratory Analysis 3: Monthly Sales Trends
Revenue stayed fairly steady (£520k–£770k) from January through August 2011, then rose sharply from September onward, peaking in November 2011 at £1,503,867 — more than double the typical monthly revenue earlier in the year. This aligns with pre-holiday season shopping. December 2011 shows a lower figure, likely because the dataset doesn't cover the full month.

Exploratory Analysis 4: Most Purchased Products (by Frequency)
"WHITE HANGING HEART T-LIGHT HOLDER" was the most frequently purchased item, appearing in 2,311 separate transactions, followed by "JUMBO BAG RED RETROSPOT" (2,109) and "REGENCY CAKESTAND 3 TIER" (2,007). Comparing this to the top-selling list by quantity shows a meaningful difference — some products (like "PAPER CRAFT, LITTLE BIRDIE") sell in huge bulk quantities but less frequently, while others (like the t-light holder) are bought often in smaller amounts, suggesting broad everyday popularity rather than bulk/wholesale demand.

Exploratory Analysis 5: Customer Purchasing Behavior
Customer spending shows two distinct patterns. Some customers, like CustomerID 14646 (£280,206 across 73 orders) and 18102 (£259,657 across 60 orders), are frequent, high-value repeat buyers — likely business or wholesale accounts. Others, like CustomerID 16446 (£168,472 from just 2 orders) and 12346 (£77,183 from a single order), show occasional but extremely large single purchases, suggesting bulk buyers rather than regular shoppers. In contrast, customers like 12748 (209 orders but only £33,053 total) represent frequent, low-value repeat shoppers, closer to typical individual retail behavior.

Task 4: Data Visualization

All visualizations were built in Power BI using the cleaned dataset (OnlineRetail_Clean), imported directly from SQL Server.

Visualization 1: Top 10 Selling Products (Bar Chart)
Title: Top 10 Selling Products. X-axis: Total Quantity Sold. Y-axis: Product Description.
Finding: A small number of products account for a large share of total units sold, showing that demand is concentrated around a few popular items rather than spread evenly across the catalogue.

Visualization 2: Revenue by Country (Pie Chart)
Title: Revenue by Country. Legend: Country. Values: Total Revenue.
Finding: The United Kingdom generated the highest revenue by a wide margin, confirming it as the business's primary market, with all other countries contributing comparatively small shares.

Visualization 3: Monthly Sales Trend (Line Chart)

Title: Monthly Sales Trend. X-axis: Month-Year. Y-axis: Total Revenue.
Finding: Sales remained fairly steady for most of the year before increasing sharply during the final quarter, with November recording the highest revenue — consistent with pre-holiday season demand.

Visualization 4: Top 10 Customers by Revenue (Column Chart)
Title: Top 10 Customers by Revenue. X-axis: CustomerID. Y-axis: Total Revenue.
Finding: A mix of purchasing patterns emerged — some customers are frequent repeat buyers, while others made only a few but very large purchases, suggesting a blend of wholesale and individual retail customers.

Visualization 5: Most Purchased Products (Treemap)
Title: Most Purchased Products. Size: Number of Transactions.
Finding: Some products appear in a high number of transactions even though they are not the highest-selling by quantity, suggesting steady day-to-day demand rather than occasional bulk buying.

Task 5: Insights

- Sales are heavily driven by a small number of high-performing products, as shown in the Top-Selling Products chart.
- The United Kingdom contributes the large majority of total revenue, making it the business's strongest market by a wide margin, as shown in the Revenue by Country chart.
- Revenue increased significantly during the last quarter of the year, with the highest sales recorded in November, as shown in the Monthly Sales Trend chart.
- Products that sell in the largest quantities are not always the products purchased most frequently — the Top-Selling Products and Most Purchased Products charts tell two different stories about demand.
- Customer purchasing behavior varies considerably, with some customers placing many small orders and others making only a few high-value purchases, as shown in the Top Customers by Revenue chart.

One-Page Summary Report

Cleaning Challenges Encountered
The dataset came with several real data quality issues. A large share of rows had no CustomerID, which I kept as NULL rather than inventing a fake ID, since these most likely represented guest checkouts. Around 1,454 rows were missing a product description, which I filled with "Unknown" to avoid losing otherwise valid sales data. The dataset also contained 5,268 duplicate rows, inconsistent country names (EIRE, RSA, USA), and a set of non-sale entries — like internal stock adjustments and administrative fees — that had to be identified and removed so they wouldn't distort the sales analysis.

Key EDA Findings

- Top-selling products differ from the most frequently purchased products.
- The United Kingdom generated the highest revenue by a wide margin.
- Sales peaked in November, driven by pre-holiday demand.

- Customer purchasing behavior varies considerably between frequent small buyers and occasional bulk buyers.
- A small number of products contribute a large share of overall sales.

Top Insights

The business relies heavily on its UK market, seasonal demand has a noticeable impact on revenue around the final quarter of the year, and customer purchasing patterns differ clearly between wholesale-style bulk buyers and regular retail customers.